

Assignment 10

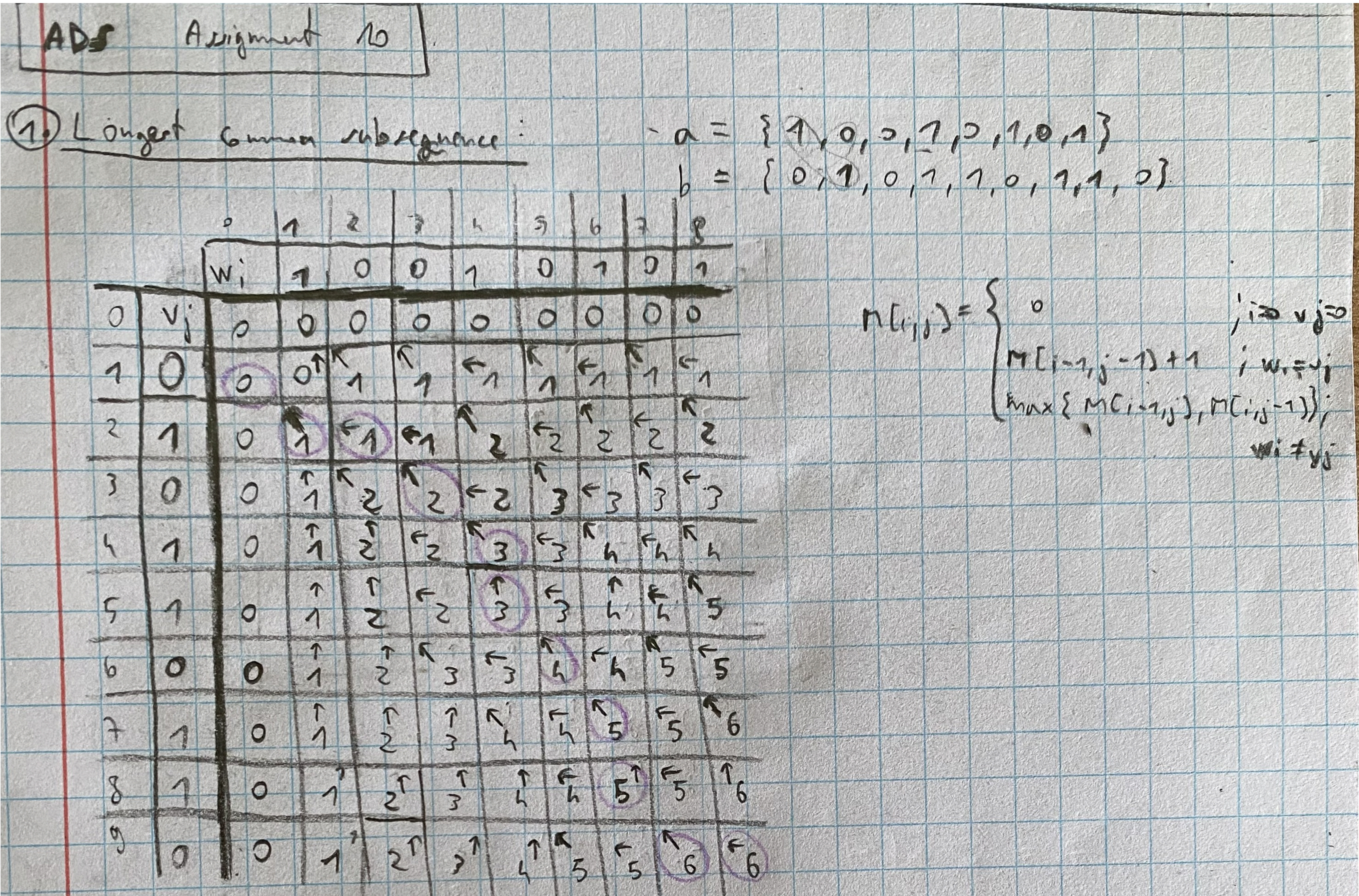
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Question 1

EXPLANATION:

I went through the matrix first row by row and then column by column. The longest common subsequence of the given strings is 6, which is shown in the matrix below:



Question 2

② n characters; string $s := "s[1], \dots, s[n]"$

Boolean function dictionary: $d(w) := \begin{cases} \text{True} & ; w \text{ is a word} \\ \text{False} & ; \text{otherwise} \end{cases}$

Can s be reconstructed?

ex. this is a word but not the whole one

$T(k-1)$ is it a word?

$T(n) = \begin{cases} \text{True} & ; j=0 \\ \{d(s[k], \dots, s[j]) \wedge T(k-1) & ; k \in [1, j]\} & ; j > 0 \end{cases}$

We can use "or" and "and"

CORRECTNESS ANALYSIS:

Above are the recurrence equations for this problem. The idea is that for every $j \in 1, \dots, n$ and for every $k \in 1, \dots, j$ we can check if the word $s[k], \dots, s[j]$ is in the dictionary and if it is, then recursively check whether there exists a solution (sequence of valid words) in $s[1], \dots, s[k-1]$. If such a solution exists, the function returns true, if it doesn't, it keeps looking until $k = j$ and until $j = n$.

In implementation, this function would run until it received the first True (and also return it) or until it traverses all possibilities, after which the function would return False.

COMPLEXITY ANALYSIS:

For every j we're checking whether the function yields true j times (because $k \in 1, \dots, j$ and because $d(w)$ takes constant time). In worst case scenario this means that complexity is $O(n^2)$.

Question 3

③ N hops

IF $i \in \{0, 1, 2, 3, 7, 8, 9\} \Rightarrow 2$ options; IF $i \in \{4, 6\} \Rightarrow 3$ options; Never 5.

$T[j] = \begin{cases} 1 & ; j=1 \\ (T[j-1] * 3 & ; p[j-1] \in \{4, 6\} \\ T[j-1] * 2 & ; p[j-1] \in \{0, 1, 2, 3, 7, 8, 9\} \end{cases}$

where p is the string of numbers we're building and T is the solution.

1 2 3 4 5 6 7 8 9

CORRECTNESS ANALYSIS:

It is equivalent whether we start from the start or from the end. The only thing we need to be careful is that if we start from the back, we need to end at key 1 (which is always the first element).

While iterating we're building a string of numbers we're currently using. That way we avoid extra calculations. Our base condition is when $N = 1$ (where our only option is number 1). At step j we need to check which number we were standing on at $(j-1)$. If $p(j-1) \in \{4, 6\}$, we multiply by 3, because you can go to 3 positions from numbers 4 and 6. If $p(j-1) \in \{0, 1, 2, 3, 7, 8, 9\}$, we multiply by 2, because you can go to 2 positions from those numbers. You cannot reach number 5 from any other number, so we've covered every case. $T[N]$ is the final result of elements we can get.

Question 4

④ * a string of letters and blanks
? a letter or blank character

Problem: Given w and p return the number of ways in which w matches p . (if it does)

$w := w[1], \dots, w[n]$
 $p := p[1], \dots, p[k]$

Recurrence Relation:

$$n[i, j] = \begin{cases} \text{false} & ; w[i] \neq p[j] \\ 0 & ; i=0 \vee j=0 \\ n[i-1, j-1] & ; w[i] = p[j] \\ n[i, j-1] \vee n[i-1, j-1] & ; p[j] = "?" \\ n[i, j-1] \vee n[i-1, j-1] \vee n[i-1, j] & ; p[j] = "*" \end{cases}$$

abababa
a*b*a

Tree diagram showing matches and mismatches:

```
graph TD
    A(abababa  
a*b*a) --> B(abab  
a*b*)
    B --> C(abab  
a*b)
    B --> D(abba  
a*b*)
    B --> E(abba  
a*b)
    E --> F(False)
    C --> G(abba  
a*)
    C --> H(abba  
a*b)
    D --> I(abba  
a*)
    D --> J(abba  
a*b*)
    D --> K(abba  
a*b)
    G --> L(abba  
a)
    G --> M(abba  
a*)
    G --> N(abba  
a)
    H --> O(abba  
a*b)
    H --> P(abba  
a*b*)
    H --> Q(abba  
a*b)
    I --> R(abba  
a*)
    I --> S(abba  
a*b)
    I --> T(abba  
a)
    J --> U(abba  
a*)
    J --> V(abba  
a*b*)
    J --> W(abba  
a*b)
    K --> X(abba  
a*b)
    K --> Y(abba  
a*b*)
    K --> Z(abba  
a*b)
    L --> AA(abba  
a)
    L --> AB(abba  
a*)
    L --> AC(abba  
a)
    M --> AD(abba  
a*)
    M --> AE(abba  
a*b)
    M --> AF(abba  
a)
    N --> AG(abba  
a)
    N --> AH(abba  
a*)
    N --> AI(abba  
a)
    O --> AJ(abba  
a*b)
    O --> AK(abba  
a*b*)
    O --> AL(abba  
a*b)
    P --> AM(abba  
a*b*)
    P --> AN(abba  
a*b*)
    P --> AO(abba  
a*b*)
    Q --> AP(abba  
a*b)
    Q --> AQ(abba  
a*b*)
    Q --> AR(abba  
a*b)
    R --> AS(abba  
a*)
    R --> AT(abba  
a*b)
    R --> AU(abba  
a)
    S --> AV(abba  
a*)
    S --> AW(abba  
a*b)
    S --> AX(abba  
a)
    T --> AY(abba  
a*)
    T --> AZ(abba  
a*b)
    T --> BA(abba  
a)
    U --> BB(abba  
a*)
    U --> BC(abba  
a*b)
    U --> BD(abba  
a)
    V --> BE(abba  
a*)
    V --> BF(abba  
a*b)
    V --> BG(abba  
a)
    W --> BH(abba  
a*b)
    W --> BI(abba  
a*b*)
    W --> BJ(abba  
a*b)
    X --> BK(abba  
a*b)
    X --> BL(abba  
a*b*)
    X --> BM(abba  
a*b)
    Y --> BN(abba  
a*b*)
    Y --> BO(abba  
a*b*)
    Y --> BP(abba  
a*b*)
    Z --> BQ(abba  
a*b)
    Z --> BR(abba  
a*b*)
    Z --> BS(abba  
a*b)
    AA --> BT(abba  
a)
    AA --> BU(abba  
a*)
    AA --> BV(abba  
a)
    AB --> BV
    AB --> BW(abba  
a*)
    AB --> BX(abba  
a)
    AC --> BY(abba  
a)
    AC --> BZ(abba  
a*)
    AC --> CA(abba  
a)
    AD --> CB(abba  
a*)
    AD --> CC(abba  
a*b)
    AD --> CD(abba  
a)
    AE --> CE(abba  
a*)
    AE --> CF(abba  
a*b)
    AE --> CG(abba  
a)
    AF --> CH(abba  
a)
    AF --> CI(abba  
a*)
    AF --> CJ(abba  
a)
    AG --> CK(abba  
a)
    AG --> CL(abba  
a*)
    AG --> CM(abba  
a)
    AH --> CN(abba  
a*)
    AH --> CO(abba  
a*b)
    AH --> CP(abba  
a)
    AI --> CQ(abba  
a)
    AI --> CR(abba  
a*)
    AI --> CS(abba  
a)
    AJ --> CT(abba  
a*b)
    AJ --> CU(abba  
a*b*)
    AJ --> CV(abba  
a*b)
    AK --> CW(abba  
a*b*)
    AK --> CX(abba  
a*b*)
    AK --> CY(abba  
a*b*)
    AL --> CZ(abba  
a*b)
    AL --> DA(abba  
a*b*)
    AL --> DB(abba  
a*b)
    AM --> DC(abba  
a*b*)
    AM --> DD(abba  
a*b*)
    AM --> DE(abba  
a*b*)
    AN --> DF(abba  
a*b*)
    AN --> DG(abba  
a*b*)
    AN --> DH(abba  
a*b*)
    AO --> DI(abba  
a*b*)
    AO --> DJ(abba  
a*b*)
    AO --> DK(abba  
a*b*)
    AP --> DL(abba  
a*b)
    AP --> DM(abba  
a*b*)
    AP --> DN(abba  
a*b)
    AQ --> DO(abba  
a*b*)
    AQ --> DP(abba  
a*b*)
    AQ --> DQ(abba  
a*b*)
    AR --> DR(abba  
a*b)
    AR --> DS(abba  
a*b*)
    AR --> DT(abba  
a*b)
    AS --> DU(abba  
a*)
    AS --> DV(abba  
a*b)
    AS --> DW(abba  
a)
    AT --> DX(abba  
a*)
    AT --> DY(abba  
a*b)
    AT --> DZ(abba  
a)
    AU --> EA(abba  
a)
    AU --> EB(abba  
a*)
    AU --> EC(abba  
a)
    AV --> ED(abba  
a*b)
    AV --> EE(abba  
a*b*)
    AV --> EF(abba  
a*b)
    AW --> EG(abba  
a*b*)
    AW --> EH(abba  
a*b*)
    AW --> EI(abba  
a*b*)
    AX --> EJ(abba  
a*b)
    AX --> EK(abba  
a*b*)
    AX --> EL(abba  
a*b)
    AY --> EM(abba  
a*b*)
    AY --> EN(abba  
a*b*)
    AY --> EO(abba  
a*b*)
    AZ --> EP(abba  
a*b)
    AZ --> EQ(abba  
a*b*)
    AZ --> ER(abba  
a*b)
    BA --> ES(abba  
a)
    BA --> ET(abba  
a*)
    BA --> EU(abba  
a)
    BB --> EV(abba  
a)
    BB --> EW(abba  
a*)
    BB --> EX(abba  
a)
    BC --> EY(abba  
a*)
    BC --> EZ(abba  
a*b)
    BC --> EA(abba  
a)
    BD --> EB
    BD --> EC
    BD --> ED
    BE --> EF
    BE --> EG
    BE --> EH
    BF --> EI
    BF --> EJ
    BF --> EK
    BG --> EL
    BG --> EM
    BG --> EN
    BH --> EO
    BH --> EP
    BH --> EQ
    BI --> ER
    BI --> ES
    BI --> ET
    BJ --> EU
    BJ --> EV
    BJ --> EW
    BK --> EX
    BK --> EY
    BK --> EZ
    BL --> FA(abba  
a*)
    BL --> FB(abba  
a*b)
    BL --> FC(abba  
a)
    BM --> FD(abba  
a*)
    BM --> FE(abba  
a*b)
    BM --> FF(abba  
a)
    BN --> FG(abba  
a*)
    BN --> FH(abba  
a*b)
    BN --> FI(abba  
a)
    BO --> FJ(abba  
a*)
    BO --> FK(abba  
a*b)
    BO --> FL(abba  
a)
    BP --> FM(abba  
a*)
    BP --> FN(abba  
a*b)
    BP --> FO(abba  
a)
    BQ --> FP(abba  
a*)
    BQ --> FQ(abba  
a*b)
    BQ --> FR(abba  
a)
    BR --> FS(abba  
a*)
    BR --> FT(abba  
a*b)
    BR --> FU(abba  
a)
    BS --> FV(abba  
a*)
    BS --> FW(abba  
a*b)
    BS --> FX(abba  
a)
    BT --> FY(abba  
a)
    BT --> FZ(abba  
a*)
    BT --> GA(abba  
a)
    BU --> GB(abba  
a*)
    BU --> GC(abba  
a)
    BU --> GD(abba  
a*)
    BV --> GE(abba  
a)
    BV --> GF(abba  
a*)
    BV --> GH(abba  
a)
    BW --> GI(abba  
a*)
    BW --> GJ(abba  
a*b)
    BW --> GK(abba  
a)
    BX --> GL(abba  
a)
    BX --> GM(abba  
a*)
    BX --> GN(abba  
a)
    BY --> GO(abba  
a)
    BY --> GP(abba  
a*)
    BY --> GQ(abba  
a)
    CA --> GR(abba  
a)
    CA --> GS(abba  
a*)
    CA --> GT(abba  
a)
    CB --> GU(abba  
a*)
    CB --> GV(abba  
a*b)
    CB --> GW(abba  
a)
    CC --> GX(abba  
a*)
    CC --> GY(abba  
a*b)
    CC --> GZ(abba  
a)
    CD --> HA(abba  
a*)
    CD --> HB(abba  
a*b)
    CD --> HC(abba  
a)
    CE --> HD(abba  
a*)
    CE --> HE(abba  
a*b)
    CE --> HF(abba  
a)
    CF --> HG(abba  
a*)
    CF --> HH(abba  
a*b)
    CF --> HI(abba  
a)
    CG --> HJ(abba  
a*)
    CG --> HK(abba  
a*b)
    CG --> HL(abba  
a)
    CH --> HM(abba  
a)
    CH --> HN(abba  
a*)
    CH --> HO(abba  
a)
    CI --> HP(abba  
a)
    CI --> HQ(abba  
a*)
    CI --> HR(abba  
a)
    CJ --> HS(abba  
a)
    CJ --> HT(abba  
a*)
    CJ --> HU(abba  
a)
    CK --> HV(abba  
a)
    CK --> HW(abba  
a*)
    CK --> HX(abba  
a)
    CL --> HY(abba  
a)
    CL --> HZ(abba  
a*)
    CL --> IA(abba  
a)
    CM --> IB(abba  
a*)
    CM --> IC(abba  
a)
    CM --> ID(abba  
a*)
    CN --> IE(abba  
a)
    CN --> IF(abba  
a*)
    CN --> IG(abba  
a)
    CO --> IH(abba  
a*)
    CO --> II(abba  
a*b)
    CO --> IO(abba  
a)
    CP --> IJ(abba  
a*)
    CP --> IK(abba  
a*b)
    CP --> IL(abba  
a)
    CQ --> IM(abba  
a*)
    CQ --> IN(abba  
a*b)
    CQ --> IO
    CR --> IP(abba  
a*)
    CR --> IQ(abba  
a*b)
    CR --> IR(abba  
a)
    CS --> IS(abba  
a*)
    CS --> IT(abba  
a*b)
    CS --> IU(abba  
a)
    CT --> IV(abba  
a*b)
    CT --> IU
    CT --> IV
    CU --> IW(abba  
a*b*)
    CU --> IX(abba  
a*b*)
    CU --> IY(abba  
a*b*)
    CV --> IZ(abba  
a*b*)
    CV --> JA(abba  
a*b*)
    CV --> JB(abba  
a*b*)
    CW --> JC(abba  
a*b*)
    CW --> JD(abba  
a*b*)
    CW --> JE(abba  
a*b*)
    CX --> JF(abba  
a*b*)
    CX --> JG(abba  
a*b*)
    CX --> JH(abba  
a*b*)
    CY --> JI(abba  
a*b*)
    CY --> JJ(abba  
a*b*)
    CY --> JK(abba  
a*b*)
    CZ --> JL(abba  
a*b*)
    CZ --> JM(abba  
a*b*)
    CZ --> JN(abba  
a*b*)
    DA --> JO(abba  
a*b*)
    DA --> JP(abba  
a*b*)
    DA --> JQ(abba  
a*b*)
    DB --> JR(abba  
a*b*)
    DB --> JS(abba  
a*b*)
    DB --> JT(abba  
a*b*)
    DC --> JU(abba  
a*b*)
    DC --> JV(abba  
a*b*)
    DC --> JW(abba  
a*b*)
    DD --> JX(abba  
a*b*)
    DD --> JY(abba  
a*b*)
    DD --> JZ(abba  
a*b*)
    DE --> KA(abba  
a*b*)
    DE --> KB(abba  
a*b*)
    DE --> KC(abba  
a*b*)
    DF --> KD(abba  
a*b*)
    DF --> KE(abba  
a*b*)
    DF --> KF(abba  
a*b*)
    DG --> KG(abba  
a*b*)
    DG --> KH(abba  
a*b*)
    DG --> KH
    DH --> KI(abba  
a*b*)
    DH --> KJ(abba  
a*b*)
    DH --> KK(abba  
a*b*)
    DI --> KL(abba  
a*b*)
    DI --> KM(abba  
a*b*)
    DI --> KN(abba  
a*b*)
    DJ --> KO(abba  
a*b*)
    DJ --> KP(abba  
a*b*)
    DJ --> KQ(abba  
a*b*)
    DK --> KR(abba  
a*b*)
    DK --> KS(abba  
a*b*)
    DK --> KT(abba  
a*b*)
    DL --> KU(abba  
a*b*)
    DL --> KV(abba  
a*b*)
    DL --> KW(abba  
a*b*)
    DM --> KX(abba  
a*b*)
    DM --> KY(abba  
a*b*)
    DM --> KZ(abba  
a*b*)
    DN --> LA(abba  
a*b*)
    DN --> LB(abba  
a*b*)
    DN --> LC(abba  
a*b*)
    DO --> LD(abba  
a*b*)
    DO --> LE(abba  
a*b*)
    DO --> LE
    DP --> LF(abba  
a*b*)
    DP --> LG(abba  
a*b*)
    DP --> LH(abba  
a*b*)
    DQ --> LI(abba  
a*b*)
    DQ --> LJ(abba  
a*b*)
    DQ --> LK(abba  
a*b*)
    DR --> LL(abba  
a*b*)
    DR --> LM(abba  
a*b*)
    DR --> LN(abba  
a*b*)
    DS --> LO(abba  
a*b*)
    DS --> LP(abba  
a*b*)
    DS --> LQ(abba  
a*b*)
    DT --> LR(abba  
a*b*)
    DT --> LS(abba  
a*b*)
    DT --> LT(abba  
a*b*)
    DU --> LU(abba  
a*b*)
    DU --> LV(abba  
a*b*)
    DU --> LV
    DV --> LW(abba  
a*b*)
    DV --> LX(abba  
a*b*)
    DV --> LY(abba  
a*b*)
    DW --> LY
    DW --> LZ(abba  
a*b*)
    DW --> MA(abba  
a*b*)
    DX --> MB(abba  
a*b*)
    DX --> MC(abba  
a*b*)
    DX --> MD(abba  
a*b*)
    DY --> ME(abba  
a*b*)
    DY --> MF(abba  
a*b*)
    DY --> MG(abba  
a*b*)
    EZ --> MH(abba  
a*b*)
    EZ --> MI(abba  
a*b*)
    EZ --> MI
    EA --> MJ(abba  
a*b*)
    EA --> MK(abba  
a*b*)
    EA --> ML(abba  
a*b*)
    EB --> MN(abba  
a*b*)
    EB --> MO(abba  
a*b*)
    EB --> MP(abba  
a*b*)
    EC --> MQ(abba  
a*b*)
    EC --> MR(abba  
a*b*)
    EC --> MS(abba  
a*b*)
    ED --> MT(abba  
a*b*)
    ED --> MU(abba  
a*b*)
    ED --> MV(abba  
a*b*)
    EE --> MW(abba  
a*b*)
    EE --> MX(abba  
a*b*)
    EE --> MY(abba  
a*b*)
    EF --> MZ(abba  
a*b*)
    EF --> NA(abba  
a*b*)
    EF --> NB(abba  
a*b*)
    EG --> NC(abba  
a*b*)
    EG --> ND(abba  
a*b*)
    EG --> NE(abba  
a*b*)
    EH --> NF(abba  
a*b*)
    EH --> NG(abba  
a*b*)
    EH --> NH(abba  
a*b*)
    EI --> NI(abba  
a*b*)
    EI --> NJ(abba  
a*b*)
    EI --> NK(abba  
a*b*)
    EJ --> NL(abba  
a*b*)
    EJ --> NM(abba  
a*b*)
    EJ --> NO(abba  
a*b*)
    EK --> NP(abba  
a*b*)
    EK --> NQ(abba  
a*b*)
    EK --> NR(abba  
a*b*)
    EL --> NS(abba  
a*b*)
    EL --> NT(abba  
a*b*)
    EL --> NU(abba  
a*b*)
    EM --> NV(abba  
a*b*)
    EM --> NW(abba  
a*b*)
    EM --> NX(abba  
a*b*)
    EN --> NY(abba  
a*b*)
    EN --> NZ(abba  
a*b*)
    EN --> NY
    EO --> OJ(abba  
a*b*)
    EO --> OK(abba  
a*b*)
    EO --> OL(abba  
a*b*)
    EP --> OM(abba  
a*b*)
    EP --> ON(abba  
a*b*)
    EP --> OP(abba  
a*b*)
    EQ --> OQ(abba  
a*b*)
    EQ --> OR(abba  
a*b*)
    EQ --> OS(abba  
a*b*)
    ER --> OT(abba  
a*b*)
    ER --> OU(abba  
a*b*)
    ER --> OV(abba  
a*b*)
    ES --> OW(abba  
a*b*)
    ES --> OX(abba  
a*b*)
    ES --> OY(abba  
a*b*)
    ET --> OZ(abba  
a*b*)
    ET --> PA(abba  
a*b*)
    ET --> PB(abba  
a*b*)
    EU --> PC(abba  
a*b*)
    EU --> PD(abba  
a*b*)
    EU --> PE(abba  
a*b*)
    EV --> PF(abba  
a*b*)
    EV --> PG(abba  
a*b*)
    EV --> PH(abba  
a*b*)
    EW --> PI(abba  
a*b*)
    EW --> PJ(abba  
a*b*)
    EW --> PK(abba  
a*b*)
    EX --> PL(abba  
a*b*)
    EX --> PM(abba  
a*b*)
    EX --> PN(abba  
a*b*)
    EY --> PO(abba  
a*b*)
    EY --> PP(abba  
a*b*)
    EY --> PQ(abba  
a*b*)
    EZ --> PR(abba  
a*b*)
    EZ --> PS(abba  
a*b*)
    EZ --> PT(abba  
a*b*)
    FA --> PU(abba  
a*b*)
    FA --> PV(abba  
a*b*)
    FA --> PW(abba  
a*b*)
    FB --> PY(abba  
a*b*)
    FB --> PZ(abba  
a*b*)
    FB --> QA(abba  
a*b*)
    FC --> QJ(abba  
a*b*)
    FC --> QK(abba  
a*b*)
    FC --> QL(abba  
a*b*)
    GD --> QM(abba  
a*b*)
    GD --> QN(abba  
a*b*)
    GD --> QO(abba  
a*b*)
    GE --> QP(abba  
a*b*)
    GE --> QQ(abba  
a*b*)
    GE --> QR(abba  
a*b*)
    HF --> QS(abba  
a*b*)
    HF --> QT(abba  
a*b*)
    HF --> QU(abba  
a*b*)
    HG --> QV(abba  
a*b*)
    HG --> QW(abba  
a*b*)
    HG --> QX(abba  
a*b*)
    HH --> QY(abba  
a*b*)
    HH --> QZ(abba  
a*b*)
    HH --> QY
    HI --> RA(abba  
a*b*)
    HI --> RB(abba  
a*b*)
    HI --> RC(abba  
a*b*)
    IJ --> RD(abba  
a*b*)
    IJ --> RE(abba  
a*b*)
    IJ --> RF(abba  
a*b*)
    IK --> RG(abba  
a*b*)
    IK --> RH(abba  
a*b*)
    IK --> RI(abba  
a*b*)
    IL --> RJ(abba  
a*b*)
    IL --> RK(abba  
a*b*)
    IL --> RL(abba  
a*b*)
    IM --> RM(abba  
a*b*)
    IM --> RN(abba  
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    IM --> RO(abba  
a*b*)
    IN --> RP(abba  
a*b*)
    IN --> RQ(abba  
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    IN --> RR(abba  
a*b*)
    IO --> RS(abba  
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    IO --> RT(abba  
a*b*)
    IO --> RU(abba  
a*b*)
    IP --> RV(abba  
a*b*)
    IP --> RW(abba  
a*b*)
    IP --> RX(abba  
a*b*)
    IQ --> RY(abba  
a*b*)
    IQ --> RZ(abba  
a*b*)
    IQ --> RY
    IR --> SA(abba  
a*b*)
    IR --> SB(abba  
a*b*)
    IR --> SC(abba  
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    IS --> SD(abba  
a*b*)
    IS --> SE(abba  
a*b*)
    IS --> SF(abba  
a*b*)
    IT --> SG(abba  
a*b*)
    IT --> SH(abba  
a*b*)
    IT --> SI(abba  
a*b*)
    IU --> SJ(abba  
a*b*)
    IU --> SK(abba  
a*b*)
    IU --> SL(abba  
a*b*)
    IV --> SM(abba  
a*b*)
    IV --> SN(abba  
a*b*)
    IV --> SO(abba  
a*b*)
    JV --> SP(abba  
a*b*)
    JV --> SQ(abba  
a*b*)
    JV --> SR(abba  
a*b*)
    JW --> SS(abba  
a*b*)
    JW --> ST(abba  
a*b*)
    JW --> SU(abba  
a*b*)
    KX --> SV(abba  
a*b*)
    KX --> SW(abba  
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    KX --> SX(abba  
a*b*)
    KY --> SY(abba  
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    KY --> SZ(abba  
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    KY --> SY
    KZ --> TA(abba  
a*b*)
    KZ --> TB(abba  
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    KZ --> TC(abba  
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    LA --> TD(abba  
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    LA --> TE(abba  
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    LA --> TF(abba  
a*b*)
    LB --> TG(abba  
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    LB --> TH(abba  
a*b*)
    LB --> TI(abba  
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    LC --> TJ(abba  
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    LC --> TK(abba  
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    LC --> TL(abba  
a*b*)
    LE --> TM(abba  
a*b*)
    LE --> TN(abba  
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    LE --> TO(abba  
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    LD --> TP(abba  
a*b*)
    LD --> TQ(abba  
a*b*)
    LD --> TR(abba  
a*b*)
    LE --> TS(abba  
a*b*)
    LE --> TT(abba  
a*b*)
    LE --> TU(abba  
a*b*)
    LF --> TV(abba  
a*b*)
    LF --> TW(abba  
a*b*)
    LF --> TX(abba  
a*b*)
    LG --> TY(abba  
a*b*)
    LG --> TZ(abba  
a*b*)
    LG --> TY
    LH --> UA(abba  
a*b*)
    LH --> UB(abba  
a*b*)
    LH --> UC(abba  
a*b*)
    LI --> UD(abba  
a*b*)
    LI --> UE(abba  
a*b*)
    LI --> UF(abba  
a*b*)
    LJ --> UG(abba  
a*b*)
    LJ --> UH(abba  
a*b*)
    LJ --> UI(abba  
a*b*)
    LK --> UJ(abba  
a*b*)
    LK --> UK(abba  
a*b*)
    LK --> UL(abba  
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    LM --> UM(abba  
a*b*)
    LM --> UN(abba  
a*b*)
    LM --> UO(abba  
a*b*)
    LN --> UP(abba  
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    LN --> UQ(abba  
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    LN --> UR(abba  
a*b*)
    LO --> US(abba  
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    LO --> UT(abba  
a*b*)
    LO --> UV(abba  
a*b*)
    LP --> UV
    LP --> UW(abba  
a*b*)
    LP --> UX(abba  
a*b*)
    LQ --> UY(abba  
a*b*)
    LQ --> UZ(abba  
a*b*)
    LQ --> UY
    LR --> VA(abba  
a*b*)
    LR --> VB(abba  
a*b*)
    LR --> VC(abba  
a*b*)
    LS --> VD(abba  
a*b*)
    LS --> VE(abba  
a*b*)
    LS --> VF(abba  
a*b*)
    LT --> VG(abba  
a*b*)
    LT --> VH(abba  
a*b*)
    LT --> VI(abba  
a*b*)
    LU --> VJ(abba  
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    LU --> VK(abba  
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    LU --> VL(abba  
a*b*)
    LV --> VM(abba  
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    LV --> VN(abba  
a*b*)
    LV --> VO(abba  
a*b*)
    LW --> VP(abba  
a*b*)
    LW --> VQ(abba  
a*b*)
    LW --> VR(abba  
a*b*)
    LX --> VS(abba  
a*b*)
    LX --> VT(abba  
a*b*)
    LX --> VU(abba  
a*b*)
    LY --> VV(abba  
a*b*)
    LY --> VW(abba  
a*b*)
    LY --> VX(abba  
a*b*)
    LZ --> VY(abba  
a*b*)
    LZ --> VZ(abba  
a*b*)
    LZ --> VY
    MA --> WA(abba  
a*b*)
    MA --> WB(abba  
a*b*)
    MA --> WC(abba  
a*b*)
    MB --> WD(abba  
a*b*)
    MB --> WE(abba  
a*b*)
    MB --> WF(abba  
a*b*)
    MC --> WG(abba  
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    MC --> WH(abba  
a*b*)
    MC --> WI(abba  
a*b*)
    MD --> WJ(abba  
a*b*)
    MD --> WK(abba  
a*b*)
    MD --> WL(abba  
a*b*)
    ME --> WM(abba  
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    ME --> WN(abba  
a*b*)
    ME --> WO(abba  
a*b*)
    MF --> WP(abba  
a*b*)
    MF --> WQ(abba  
a*b*)
    MF --> WR(abba  
a*b*)
    MG --> WS(abba  
a*b*)
    MG --> WT(abba  
a*b*)
    MG --> WU(abba  
a*b*)
    MH --> WV(abba  
a*b*)
    MH --> WX(abba  
a*b*)
    MH --> WY(abba  
a*b*)
    MI --> WZ(abba  
a*b*)
    MI --> XA(abba  
a*b*)
    MI --> XB(abba  
a*b*)
    MJ --&
```